

AVIGNON PROVENCE AP, France

WMO: 075630

Lat: 43.9114N

Lon: 4.9025E

Elev: 157

StdP: 14.61

Time Zone: 1.00 (EUC)

Period: 05-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	25.3	28.0	14.6	11.6	33.6	19.0	14.4	35.0	38.0	41.6	33.8	42.4	6.4	350	0.511

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	23.9	94.6	70.7	92.0	69.4	89.4	68.8	73.4	88.8	71.9	86.6	70.6	84.6	9.0	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
68.7	106.0	78.6	67.2	100.7	77.7	65.8	95.7	76.8	37.0	88.6	35.6	86.6	34.5	84.4	78.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 30.6	26.8	23.5	DB	21.1	99.6	3.2	3.3	18.8	102.0	16.9	103.9	15.1	105.8	12.8	108.1	(4)
(5)			WB	19.9	76.0	3.7	1.5	17.3	77.1	15.1	77.9	13.1	78.8	10.4	79.8	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	58.9	43.0	44.3	50.7	57.2	63.5	71.9	77.1	75.2	68.3	60.6	50.6	43.8	(6)	
	DBStd	13.19	6.06	6.32	5.70	5.12	5.30	5.34	4.27	4.37	4.98	5.95	6.74	6.34	(7)	
	HDD50	752	229	173	59	5	0	0	0	0	0	4	73	209	(8)	
	HDD65	3307	681	580	442	237	91	6	0	0	24	157	433	656	(9)	
	CDD50	4018	12	13	82	221	418	658	840	782	550	334	90	18	(10)	
	CDD65	1099	0	0	0	4	43	214	375	318	123	22	0	0	(11)	
	CDH74	11092	0	0	5	89	472	2228	4020	3131	1033	114	0	0	(12)	
	CDH80	4646	0	0	0	6	100	910	1929	1395	299	7	0	0	(13)	
(14) Wind	WSAvg	9.0	9.6	10.7	9.6	8.9	9.4	8.5	8.8	8.6	8.1	8.4	8.9	8.5	(14)	
(15) Precipitation	PrecAvg	26.2	2.1	1.4	1.3	2.4	1.9	1.5	1.1	1.4	4.0	3.5	3.6	1.9	(15)	
	PrecMax	39.0	6.0	5.0	3.5	6.2	4.8	4.1	4.2	3.9	10.1	6.5	9.9	7.3	(16)	
	PrecMin	16.4	0.0	0.0	0.0	0.2	0.2	0.1	0.0	0.3	1.1	1.2	0.7	0.0	(17)	
	PrecStd	6.2	1.6	1.3	1.1	1.4	1.2	1.1	1.1	0.9	2.3	1.6	2.4	1.7	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	63.4	65.8	74.2	80.6	87.1	96.7	97.6	98.3	91.0	80.4	69.1	62.5	(19)	
		MCWB	53.9	53.1	55.4	62.1	65.8	72.2	71.8	71.4	68.5	65.0	59.0	55.9	(20)	
	2%	DB	59.2	61.5	68.8	76.7	82.7	92.1	94.7	94.3	86.8	77.1	66.5	59.8	(21)	
		MCWB	51.3	50.9	54.8	60.6	63.9	69.8	71.0	70.3	67.3	64.7	58.8	53.7	(22)	
	5%	DB	56.5	58.2	65.7	73.0	79.1	89.0	92.4	91.0	83.4	74.4	64.0	57.4	(23)	
		MCWB	50.2	49.2	53.3	58.7	62.5	68.7	69.5	69.5	66.2	63.4	57.3	52.1	(24)	
	10%	DB	53.7	55.3	62.5	69.4	75.9	85.6	90.0	87.8	80.3	71.6	61.4	54.8	(25)	
		MCWB	48.0	48.0	51.7	56.7	61.1	67.5	68.9	68.3	65.3	61.9	55.7	49.9	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	54.8	54.2	57.6	63.9	67.7	74.9	75.2	74.5	72.0	68.1	61.3	57.2	(27)	
		MCDB	61.1	62.8	67.7	77.4	81.5	93.9	90.7	91.0	83.0	76.4	66.3	61.2	(28)	
	2%	WB	52.7	52.2	55.9	61.6	65.7	72.0	73.6	73.0	70.0	66.4	59.5	54.7	(29)	
		MCDB	58.1	58.9	66.0	74.3	78.8	87.8	88.9	88.2	80.8	74.5	64.8	58.8	(30)	
	5%	WB	50.7	50.5	54.5	59.4	64.1	70.1	72.1	71.5	68.3	64.8	58.2	52.6	(31)	
		MCDB	55.7	56.8	63.7	70.7	75.9	85.1	86.9	85.8	78.8	72.5	63.2	56.6	(32)	
	10%	WB	48.6	48.5	52.8	57.5	62.5	68.5	70.7	70.1	66.7	62.9	56.6	50.4	(33)	
		MCDB	53.0	54.5	60.9	67.4	73.4	82.5	85.1	83.5	77.0	69.9	60.8	54.1	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	15.7	17.4	20.1	21.3	21.2	22.9	23.9	23.6	21.7	18.5	15.9	15.4	(35)	
		MCDBR	19.3	23.4	26.1	28.0	26.8	28.6	28.3	29.3	27.1	22.0	19.2	18.2	(36)	
	5% WB	MCWBR	12.8	14.6	14.4	14.4	12.1	11.0	9.7	10.6	11.0	11.5	12.5	12.7	(37)	
		MCWBR	17.3	21.1	23.1	25.7	24.7	27.0	26.1	26.5	23.2	19.7	17.1	15.9	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.323	0.343	0.382	0.398	0.407	0.407	0.399	0.397	0.386	0.388	0.350	0.320	(40)	
		taud	2.454	2.394	2.299	2.281	2.303	2.332	2.353	2.364	2.385	2.372	2.430	2.472	(41)	
	Ebn at Noon		254	268	270	275	275	274	275	272	266	247	240	243	(42)	
		Edh at Noon	23	29	36	40	40	39	38	36	33	30	24	21	(43)	
(44) All-Sky Solar Radiation	RadAvg		541	844	1256	1635	1966	2284	2282	1958	1490	937	594	469	(44)	
(45)	RadStd		58	84	105	129	143	104	105	72	89	88	71	50	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (33 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page